| | Document No. | Section No. |
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| | Effective Date: | Revision No. |
| | **PROCEDURAL MANUAL** | |
| | **SPILL PREVENTION AND RESPONSE** | |

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| **OBJECTIVE** | This procedure outlines methods for preventing accidental spills of hazardous liquids (including fuels, oils, lubricants, solvents, paints, chemicals) on BHBI construction sites and defines the steps for a safe, prompt, and effective response should a spill occur. The aims are to protect the health and safety of personnel, minimize environmental contamination (soil, water, air), and ensure compliance with relevant regulations (DENR, local ordinances) |
| **SCOPE** | This procedure applies to all activities involving the storage, handling, use, and transportation of potentially hazardous liquids on BHBI project sites. It covers response actions for minor and moderate spills manageable with on-site resources. Major spills may require external emergency services. |

**Responsibilities**

- **Project Manager:** Overall responsibility for ensuring project compliance with environmental regulations, including those pertaining to spill prevention, hazardous waste management, and reporting. Ensures allocation of adequate resources.
- **Project Engineer:** Ensures spill prevention and hazardous waste management requirements, compliant with DENR regulations and ECC conditions, are integrated into site planning and execution. Oversees overall compliance.
- **Site Engineer:** Responsible for the day-to-day implementation of spill prevention measures, ensuring DENR-compliant hazardous waste storage, maintaining spill kits, coordinating initial spill response, and ensuring proper segregation and handling of spill debris.
- **Safety Officer / Pollution Control Officer (PCO):** Acts as the designated PCO responsible for ensuring project compliance with environmental laws (RA 6969, RA 9275, RA 8749, PD 1586/ECC conditions, etc.) and relevant DENR Administrative Orders (DAOs). Specific duties related to this procedure include:
    - Developing/updating this procedure to reflect current DENR regulations.
    - Ensuring spill prevention measures are adequately implemented.
    - Ensuring spill kits are properly stocked, located, and maintained.
    - Providing relevant environmental training (spill response, hazardous waste handling).
    - Leading significant spill response efforts.
    - Overseeing the management of hazardous waste generated (including spill debris) according to DAO 2013-22, ensuring proper segregation, storage, labeling, and use of the manifest system.
    - Preparing and submitting required environmental reports to DENR-EMB (e.g., Self-Monitoring Reports (SMRs) which may include waste generation data, spill incident reports).
    - Acting as the primary liaison with DENR-EMB Region IV-A (Calabarzon) regarding spill incidents, hazardous waste management compliance, and other pertinent environmental matters.
    - Conducting spill investigations and managing corrective actions.
    - Initiating required **DENR reporting** for significant spills/releases.
- **Foremen:** Ensures crews handle liquids correctly, understand DENR-compliant waste segregation (especially spill debris), know spill kit locations, and report spills immediately.
- **Warehouseman:** Ensures proper storage and DENR-compliant secondary containment for liquids and hazardous materials/wastes stored in designated areas. Maintains spill kits and storage records.
- **Procurement:** Ensures engagement of **DENR-accredited** transporters and Treatment, Storage, Disposal (TSD) facilities for hazardous waste generated from spills.
- **Equipment Operators/Mechanics:** Handles fuels/lubricants safely, performs leak checks, reports spills/leaks immediately.
- **All Site Personnel:** Handles liquids safely, understands basic spill prevention and response, knows spill kit locations, practices proper waste segregation, and reports spills immediately.

**Procedure**

**Part A: Spill Prevention (Aligned with RA 6969 & Good Practice)**

**4.1. Proper Storage:**

- **Designated & Compliant Areas: Store hazardous liquids and substances only in designated, secure areas compliant with RA 6969/DAO requirements – away from drains, waterways, incompatible materials, and ignition sources.**
- **Secondary Containment: Provide impervious secondary containment (e.g., spill pallets, bunded concrete areas) for bulk storage, meeting DENR standards (typically 110% of the largest container or 10% of total volume).**
- **Containers & Labeling: Use appropriate, non-leaking containers. Ensure containers are clearly labeled according to DENR requirements (chemical name, hazards) and GHS (Globally Harmonized System) standards, if applicable. Keep containers closed.**
- **Inventory & MSDS: Maintain an up-to-date inventory of hazardous substances and ensure Material Safety Data Sheets (MSDS)/Safety Data Sheets (SDS) are readily accessible, as required by DENR.**

**4.2. Proper Handling and Transfer:**

- **Safe Practices: Utilize drip pans, appropriate pumps/funnels, and designated transfer areas to minimize spills during handling and transfer operations.**
- **Fueling: Conduct fueling in designated areas with spill kits present. Prevent fuel spills from entering drains or soil.**

**4.3. Equipment Maintenance:**

- **Leak Prevention: Regularly inspect equipment for leaks as per the *Preventive Maintenance Procedure*. Repair leaks promptly.**

**4.4. Site Housekeeping:**

- **Cleanliness: Maintain clean work areas to easily identify leaks/spills.**

**4.5. Spill Kits:**

- **Availability & Contents: Ensure adequate, DENR-appropriate spill kits (containing PPE, absorbents, containment tools, disposal bags) are available, inspected regularly, and strategically located.**
    - **Deliverables: Spill Kit Inspection Log**

**4.6. Training:**

- **DENR Compliance Awareness: Train personnel on DENR requirements related to hazardous substance handling, spill prevention, response actions, and hazardous waste segregation/management during induction and toolbox talks.**

- **Deliverables: Training Records**

**Part B: Spill Response (Aligned with DENR Contingency & Reporting)**

**4.7. Initial Actions (Discoverer):**

- **SAFETY FIRST: Assess risk.**
- **STOP: Stop the source *only if safe*.**
- **ALERT: Alert supervisor (Foreman/Site Engineer/Safety Officer).**
- **SECURE: Secure area, prevent spread. Crucially, prevent spill from entering any drainage system, water body, or migrating off-site, as this constitutes a violation of RA 9275 (Clean Water Act) and likely triggers DENR reporting.**

**4.8. Notification and Assessment:**

- **Internal: Foreman/Site Engineer notifies Safety Officer/Project Engineer immediately.**
- **Assessment: Safety Officer assesses spill (material, quantity, hazards, potential to reach environment/off-site).**
- **DENR Reporting Determination: The Safety Officer, in consultation with Project Engineer/Manager, determines if the spill constitutes a "significant spill" requiring reporting to DENR-EMB Region IV-A (Calabarzon) based on:**
  - **ECC Conditions: Specific reporting thresholds defined in the project's ECC (if applicable).**
  - **Regulatory Requirements: Potential requirements under RA 6969 or RA 9275 for releases impacting the environment. Generally, spills that contaminate water bodies, migrate off-site, exceed certain volumes (if defined), or pose significant health risks require reporting. When in doubt, consult with management and potentially DENR.**
  - **Local Ordinances: Check Indang LGU requirements.**

**4.9. Containment:**

- **Prevent Spread: Use spill kit materials to contain the spill and prevent entry into drains, soil, or water bodies. Prioritize protecting sensitive areas.**

**4.10. Cleanup:**

- **PPE: Use appropriate PPE based on MSDS/SDS.**
- **Absorb: Clean up the spill using appropriate absorbents.**

**4.11. Waste Disposal (Aligned with RA 6969 / DAO 2013-22):**

- **Hazardous Waste: Contaminated soil, absorbents, rags, and PPE from spills of hazardous substances/oils are considered Hazardous Waste under RA 6969.**
- **Segregation & Containerization: Place spill debris into designated, properly labeled (Hazardous Waste details, Generator Name: BHBI-[Project Name], Address, Waste Code if known), sealed containers.**
- **Generator ID: Ensure the BHBI project is registered as a Hazardous Waste Generator with DENR-EMB if generating such waste.**
- **Storage: Store generated hazardous waste in a designated, compliant temporary storage area (secure, contained, labeled) for a period not exceeding the DENR limit (typically 1 year, subject to storage type/approval).**
- **Transport & Disposal: Engage DENR-EMB accredited Hazardous Waste Transporter and DENR-EMB accredited TSD Facility. Utilize the DENR Manifest System for tracking the waste from generator to TSD facility. Retain copies of all manifests. Refer to *Waste Management Procedure* for detailed steps.**
  - **Deliverables: Copies of DENR Hazardous Waste Manifests, Certificates of Treatment/Disposal from TSD Facility.**

**4.12. Decontamination:**

- **Clean Surfaces/Tools: Decontaminate appropriately.**

**4.13. Restock Spill Kit:**

- **Maintain Readiness: Restock used kits immediately.**

**4.14. Investigation and Reporting:**

- **Internal Report: Prepare a detailed Spill Incident Report including cause analysis and corrective actions.**
  - **Deliverables: Spill Incident Report**
- **External Reporting (If Required): If deemed a reportable incident (see 4.8), the Safety Officer/Project Manager submits the required notification/report to DENR-EMB Region IV-A within the timeframe stipulated by regulations or ECC conditions. Maintain copies of all regulatory correspondence.**
  - **Deliverables: Copies of DENR Notifications/Reports (if applicable)**

**Part C: Emergency Preparedness (Aligned with DENR Contingency Planning)**

**4.15. Emergency Contacts:**

- **Include DENR: Maintain updated emergency contact list including internal responders, external agencies, and the DENR-EMB Region IV-A hotline/contact person.**

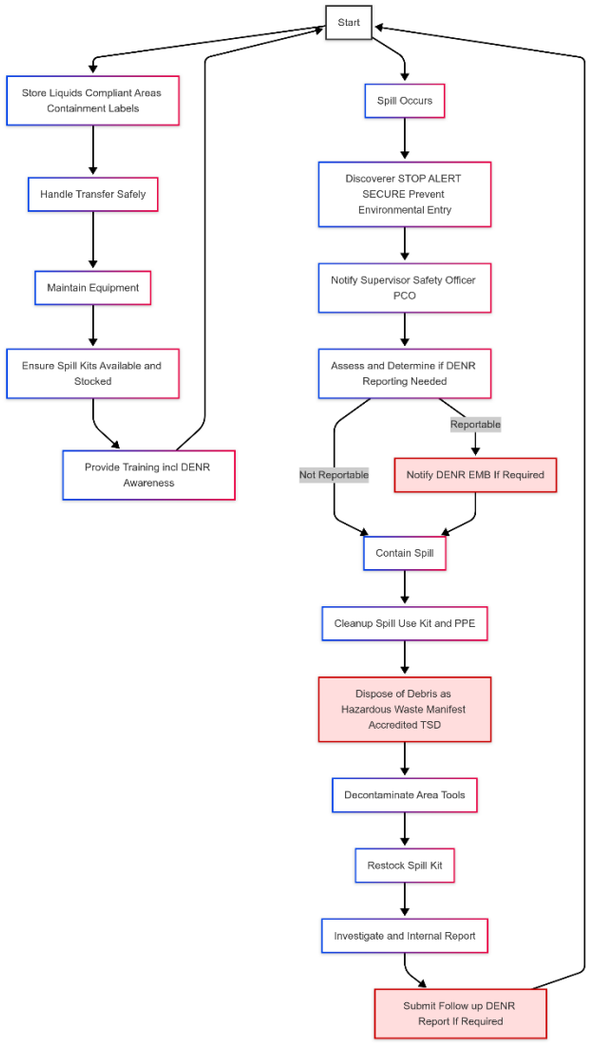
**4.16. MSDS/SDS Availability:**

- **Accessibility: Ensure easy access to MSDS/SDS for all hazardous substances.**

**Continuous Improvement & Disclaimer**

- Review incidents and update procedure based on lessons learned and regulatory changes.
- **Disclaimer:** This procedure provides general guidance. Specific requirements may vary based on project-specific ECC conditions, local ordinances, and the most current DENR regulations (RA 6969, RA 9275, associated DAOs like DAO 2013-22, DAO 2016-08, etc.). BHBI personnel must consult the latest applicable regulations and project-specific requirements.

**Flowchart**




**Records Generated (Deliverables)**

- Spill Kit Inspection Log
- Training Records (Spill Prevention, Response, HazWaste Handling)
- Spill Incident Report
- DENR Hazardous Waste Generator Registration (If applicable)
- Copies of DENR Hazardous Waste Manifests
- Certificates of Treatment/Disposal (From accredited TSD Facility)
- Copies of DENR Notifications/Reports (For reportable spills)
- MSDS/SDS Inventory/Files

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